

Fig. 3.4. Burning blighted corn. *[From* Interstellar*, used courtesy of Warner Bros. Entertainment Inc.]*

The gravitational anomalies that occur in *Interstellar* (Chapters 24 and 25), for example, the coin Cooper tosses that suddenly plunges to the floor, are *speculations*. So is harnessing the anomalies to lift colonies off Earth (Chapter 31).

Although experimental physicists when measuring gravity have searched hard for anomalies—behaviors that cannot be explained by the Newtonian or relativistic laws—no convincing gravitational anomalies have ever been seen on Earth.

However, it seems likely from the quest to understand quantum gravity that our universe is a membrane (physicists call it a “brane”) residing in a higher-dimensional “hyperspace” to which physicists give the name “bulk”; see Figure 3.5 and Chapters 4 and 21. When physicists carry Einstein’s relativistic laws into this bulk, as Professor Brand does on the blackboard in his office (Figure 3.6), they discover the possibility